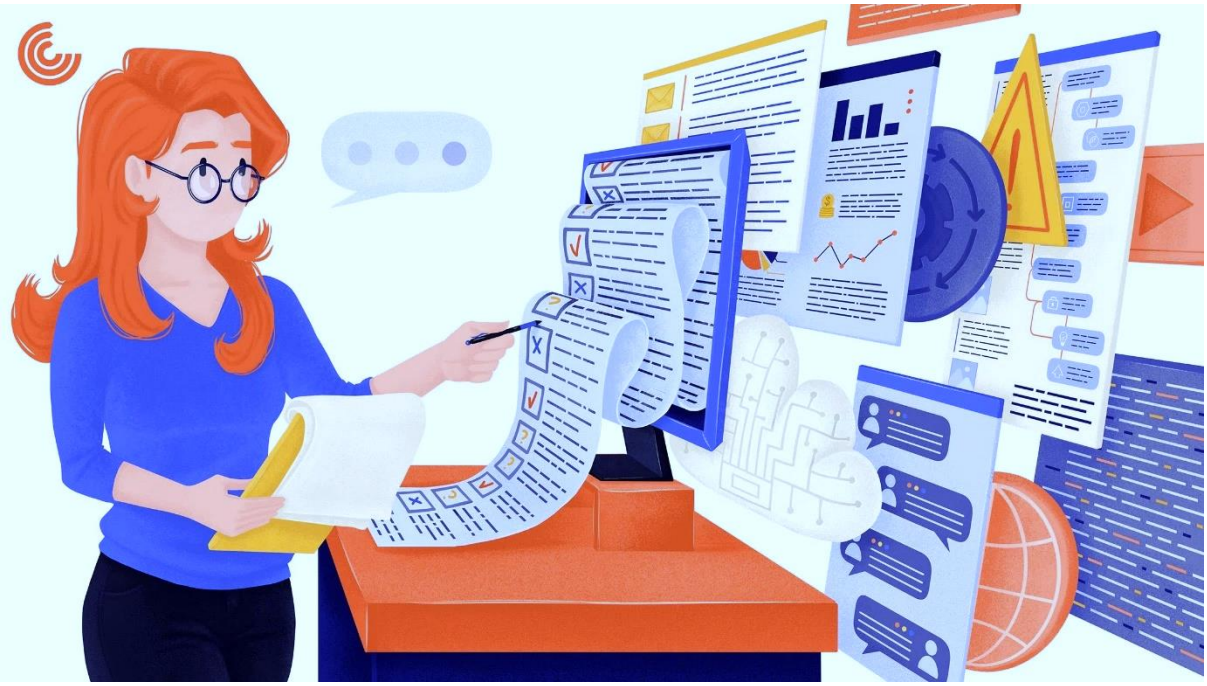
Requirement Specification Document



RAPID APPLICATION DEVELOPMENT (IT2053)

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INDEX NO: D/BIS/23/0008

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1. INTRODUCTION

The Library Management System is being developed for ABC Institute to facilitate the borrowing and returning of e-books through a user-friendly interface. This document outlines the functional and non-functional requirements for the system, using the concepts of Rapid Application Development (RAD) and implemented using Visual Studio, language C#.

2. REQUIREMENTS

I. FUNCTIONAL REQUIREMENT

a. Create a new user account

- The new user shall create a new account using a username and a unique password.
- Instruct the password (How many characters? What to include? etc.)

b. User login

- Here a user shall be able to login to the system by entering their unique UserID and password
- If the entered UserID or password is given, the system shall generate the message "password or username incorrect.
- The system should allow users to reset their passwords in case they forget them, following a secure verification process.

c. E-book Borrowing

- Users shall be able to borrow e-books from the library management system.
- Users shall be able to search for e-books by title, author, or category.
- The system shall display the status of e-books.
- The system shall check if the e-book is available for borrowing and if the user has not exceeded the borrowing limit.
- Users should receive notifications if the wanted e-book is unavailable.

d. Fine Calculation for Late Returns

- The system shall calculate fines for late book returns and update the user's account (Per late day the library charges 5/=)
- The system shall generate a calculation of the fine by the return date, due date, User ID
- The fine calculation should consider holidays and weekends that have been specified in the system.
- The fine calculation should only apply to books returned after the due date The system should validate that the return date is after the due date to apply fines.
- The system shall provide a calculation of the fine by the return date, due date, and User ID.

II. NON-FUNCTIONAL REQUIREMENT

a. Performance

- The system should provide quick response times to user actions such as searching for e-books, borrowing, returning, and logging in. Users should not experience significant delays or waiting times when interacting with the system.
- Database queries should be structured and optimized to minimize execution time and resource consumption. Proper indexing, query optimization techniques, and database design principles should be employed to ensure efficient retrieval and manipulation of data.

b. Reliability

- The system shall be available 24/7.
- The system should have strong error handling to handle unexpected failures.

c. Usability

- The user interface shall be intuitive and user-friendly, facilitating ease of navigation.
- Error messages shall be clear and informative, guiding users on how to resolve issues.
- The system should provide multilingual support for users from diverse linguistic backgrounds.

3. CONCLUSION

In conclusion, the development of the Library Management System for ABC Institute possibilities to develop the institution's method of resource management and user engagement. Through the implementation of functionality such as creating new accounts, login mechanisms, e-book borrowing facility, and fine calculation for late returns, the system ensures continuous access to digital resources while providing reliable and efficient library processes. Supported by a focus on non-functional requirements such as performance optimization, reliability, and usability, the system aims to deliver a user-friendly experience. With its commitment to excellence in both functionality and user experience, the Library Management System assured to become an essential tool in the academic excellence and knowledge distribution at ABC Institute.